

High-band coupling at $L = 0.3$: the operator-norm budget is false, the matrix Schur is the remainder PRIME.RDAGGER.KAPPA_HIGH.01 — round 480

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Abstract

Round 479 reduced $\lambda_*(0.3) \geq c$ to a single operator-norm bound: $\kappa_{\text{high}} = \|P_5 A_+ (1 - P_5)\| \leq 1.79 \cdot 10^{-2}$ at the split $(t_*, N) = (8, 5)$, with $A_+ = \sigma_A 1_{|t| \geq 8}$ and P_5 the projection on the first five Slepian modes of band 8 on $L^2[-0.3, 0.3]$. Nyström-stable quadrature kills that inequality: $\kappa_{\text{high}} \approx 2.91 \cdot 10^{-1}$ (sixteen times the budget), and every scanned rebalance of (t_*, N) stays in $[0.29, 0.40]$. The crude test compares the edge-mode singular value $\psi_4 \leftrightarrow \psi_6$ to the ground-mode eigenvalue — structurally the wrong pairing. The even 3×3 matrix Schur $S = B - CC^\top / c_{\text{tail}}$ is numerically positive ($\lambda_{\min}(S) \approx 5.69 \cdot 10^{-3}$ at $n = 48$, $T = 150$; stable versus $n = 64$). Proving $S \succ 0$ with an interval Fourier transform on $[8, T]$ and a $1/t^2$ integration-by-parts tail would close $\lambda_*(0.3) \geq c$. That enclosure is open. Verdict REDUCED(refined-Schur; crude- κ_{high} -killed@0.30). The Fourier–HS majorant of r478 is not reused. $L = 0.3$ is inside the Yoshida–Bombieri zone. No RH claim.

Status. REDUCED(refined-Schur; crude- κ_{high} -killed@0.30). No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq 0$. The named operator-norm remainder of r479 is false. The refined matrix Schur is a smaller open statement, not a certified positivity. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ the node set is empty, $P \equiv 0$. Plancherel $\hat{h}(t) = \int h(x) e^{itx} dx$, $\|h\|_2^2 = (2\pi)^{-1} \int |\hat{h}|^2$. Symbol (r476): $\sigma_A(t) = \text{Re } \psi(\frac{1}{4} + it/2) - \log \pi$, even, increasing, $\sigma_A(0) \in [-5.372183419225667, -5.372183419225665]$. The r479 contract is in force: t_c , $\text{tr}(K_{t_c})$, c_{tail} , and κ_{low} stay as sealed.

2 The named bound, and why it fails

Write $K_8 = P_L B_8 P_L$ and let $\{\psi_n\}$ be its Slepian basis on $L^2[-0.3, 0.3]$. Then K_8 is diagonal, so the low-band piece of $P_5 A (1 - P_5)$ obeys $\kappa_{\text{low}} \leq |\sigma_A(0)| \sqrt{\varepsilon}$ with $\varepsilon = \lambda_6(K_8) \leq 1.4335 \cdot 10^{-6}$ (Nyström-stable, r479). The high-band piece is

$$\kappa_{\text{high}} = \|P_5 A_+ (1 - P_5)\|, \quad A_+ = \sigma_A 1_{|t| \geq 8}.$$

In matrix form this is the 2-norm of the off-block $C_{mn} = (2\pi)^{-1} \int_{|t| \geq 8} \sigma_A(t) \widehat{\psi}_m(t) \overline{\widehat{\psi}_n(t)} dt$ for $m < 5 \leq n$.

Theorem 1 (Kill of the operator-norm budget). *At $(t_*, N) = (8, 5)$, Nyström $n = 48$ with trapezoid quadrature on $[8, 150]$ gives*

$$\kappa_{\text{high}} \approx 2.905 \cdot 10^{-1} \not\leq 1.79 \cdot 10^{-2}.$$

The same floor $\kappa_{\text{high}} > 0.25$ holds at $(8, 6)$, $(10, 6)$ and $(12, 7)$. $n = 64$ moves the value by less than $5 \cdot 10^{-3}$. Even-odd entries of A_+ vanish ($A_+[4, 5] \sim 10^{-12}$). The large entries are same-parity edge couplings $A_+[3, 5] \approx 0.213$, $A_+[4, 6] \approx 0.162$.

Modes $n = 3, 4$ of band 8 have high-band mass $1 - \lambda_n = O(1)$. The tail $n \geq 5$ is equally high-frequency. Cauchy–Schwarz on $\langle \psi_m, \sigma_A 1_{\geq 8v} \rangle$ with both sides of mass $\Theta(1)$ produces an $O(\sigma_A)$ coupling, in the same range as the measured singular value. Raising t_* or N moves the edge but does not remove it: the time-bandwidth transition always sits between the block and the tail.

The r479 budget $\sqrt{\mu_3 c_{\text{tail}}} - \kappa_{\text{low}} \leq 1.79 \cdot 10^{-2}$ compares that edge singular value to the even Fourier Higham floor $\mu_3 \geq 2.457 \cdot 10^{-3}$. Those two numbers live on different directions (the ground prolate versus $\psi_4 \leftrightarrow \psi_6$). The comparison is not forced, and the inequality is false.

The r478 Fourier–HS majorant ($\text{HS} \sim 0.27$) is a different operator and is not reused.

3 Slepian structure and the $1/t$ tail

On the low band, $1_{|t| < 8} = B_8$ and $P_L B_8 P_L = K_8$ is diagonal, so $P_5 \leftrightarrow (1 - P_5)$ through K_8 is exactly zero. All coupling of A_+ is weighted high-band overlap. The Fourier transform of a time-limited prolate does *not* decay super-exponentially: $\psi_n(\pm L) \neq 0$, and one integration by parts gives

$$|\hat{\psi}_n(t)| \leq \frac{|\psi_n(L)| + |\psi_n(-L)|}{|t|} + \frac{\|\psi'_n\|_1}{|t|}.$$

A second integration produces a $1/t^2$ majorant with explicit endpoint and $\|\psi''_n\|_1$ terms. Then $\int_T^\infty (\log t)/t^2 dt = (1 + \log T)/T < \infty$, so a high-frequency remainder is possible in principle. At $T = 150$ the elementary integral equals $4.01 \cdot 10^{-2}$. Endpoint values of ψ_0 are Nyström-stable ($n = 48$ versus $n = 64$ differ by < 0.02). $n = 16$ — sufficient for the r479 eigenvalues — is unusable for Fourier transforms: the constant-mode Rayleigh $Q(\psi_0)$ jumps by more than 1. Quadrature of $\hat{\psi}_n(t) = \sum_i w_i \psi_n(x_i) e^{itx_i}$ dies once t outruns the Gauss nodes ($T = 400$ at $n = 48$ is garbage; the working window is $T \leq 150$).

4 The refined remainder

Let B be the even 3×3 Gram of Q_W on $\text{span}\{\psi_0, \psi_2, \psi_4\}$ and C the coupling of that block to the even tail $\{\psi_6, \psi_8, \dots\}$. On the K -tail, r479 gives $A \geq c_{\text{tail}} \geq 0.2408$. The block-matrix $\begin{pmatrix} B & C \\ C^\top & T \end{pmatrix}$ is positive definite once $B \succ 0$, $T \succeq c_{\text{tail}} I$, and

$$S = B - \frac{CC^\top}{c_{\text{tail}}} \succ 0.$$

The large entries of C sit in the ψ_4 row, where $B_{44} \approx 1.3$; the ground row of C is $O(10^{-2})$.

Theorem 2 (Numeric matrix Schur). *At $(8, 5)$, $n = 48$, $T = 150$:*

$$\lambda_{\min}(B) \approx 6.66 \cdot 10^{-3}, \quad \|C\| \approx 0.206, \quad \lambda_{\min}(S) \approx 5.69 \cdot 10^{-3}.$$

The odd sector has $\lambda_{\min}(S) \approx 0.207$. $n = 64$ changes $\lambda_{\min}(S)$ by less than $5 \cdot 10^{-4}$. These are working constants, not a Lipschitz-interval enclosure of either the prolates or the Fourier integrals.

If $S \succ 0$ is proved with a quadrature remainder on $[8, T]$ and the integration-by-parts tail on $[T, \infty)$, then $\lambda_*(0.3) \geq c$ for an explicit $c > 0$ (odd sector strictly easier; Π is rank one per parity). That is the exact remainder after this round.

5 Sealed balance

Probe `kappa_high_probe.py`, SPEC_SHA 7be2cb1dcc994ba6. Smoke 14/14 in 0.05 s.

object	status	constant
κ_{high} at $(8, 5)$	KILL (numeric)	$\approx 2.91 \cdot 10^{-1}$
r479 budget	dead	$1.79 \cdot 10^{-2}$
rebalance (t_*, N)	KILL	$\kappa_{\text{high}} > 0.25$
Fourier–HS	already KILL	not reused
$c_{\text{tail}}(8, 5)$	sealed (r479)	≥ 0.2408
κ_{low}	PROVED (r479)	$\leq 6.43 \cdot 10^{-3}$
even $S = B - CC^\top / c$	numeric PD	$\lambda_{\min} \approx 5.69 \cdot 10^{-3}$
odd S	numeric PD	$\lambda_{\min} \approx 0.207$
$n = 16$ FTs	unusable	$Q(\psi_0)$ jumps by > 1
$1/t^2$ tail formula	elementary	$(1 + \log T)/T$
interval $S \succ 0$	OPEN	the remainder

6 Kill table

gate	status	note
$\kappa_{\text{high}} \leq 1.79 \cdot 10^{-2}$	KILL	$0.291 > 0.018$
rebalance the split	KILL	edge mode persists
Fourier–HS majorant	KILL	r478; not reused
crude $\sqrt{\mu_3 c} - \kappa_{\text{low}}$	KILL	mixes directions
$n = 16$ FT quadrature	KILL	evals yes, hats no
$T = 400$ trapezoid	KILL	nodes undersample e^{itx}
refined even Schur	OPEN	numeric $5.7 \cdot 10^{-3}$
scramble / anti-list	PASS	$P = 0$ vacuous; e^{2L} ; no zeros
no RH / not λ_*	PASS	type is REDUCED

7 What this does not do

It does not prove $\lambda_*(0.3) \geq 0$. It does not enclose Slepian eigenfunctions or their Fourier transforms in a Lipschitz interval. It does not treat $L > 0.3$. No cofinal quantifier. No RH claim.